

# ASSIGNMENT

Date \_\_\_\_\_ 20 \_\_\_\_\_

## TRANSPORTATION MODEL Least Cost Method

Q) Sunray Transport company ships truckloads of grains from silos to ~~four~~ mills. The supply in truckload on the different route are summarized in table. The unit of transportation cost  $C_{ij}$  (shown in the north-west corner of each box) are in hundred of dollars. The model seeks the minimum cost shipping schedule b/w the silos and the mills.

|        |   | Mills |    |    |    | Supply |
|--------|---|-------|----|----|----|--------|
|        |   | 1     | 2  | 3  | 4  |        |
| Silos  | 1 | 10    | 2  | 20 | 11 | 15     |
|        | 2 | 12    | 7  | 9  | 20 | 25     |
|        | 3 | 4     | 14 | 16 | 18 | 10     |
| Demand |   | 5     | 15 | 15 | 15 |        |

Teacher's Signature \_\_\_\_\_

AM

Page No.

Solution:

| Silos | Mills |          |    |    | Supply |
|-------|-------|----------|----|----|--------|
|       | 1     | 2        | 3  | 4  |        |
|       |       | Start 15 |    |    |        |
| 1     | 10    | 2        | 20 | 11 | 15     |
| 2     | 12    | 7        | 9  | 20 | 28     |
| 3     | 4     | 14       | 16 | 18 | 10     |

Demand ~~5~~ ~~15~~ ~~15~~ ~~15~~

The arrows show the order in which allocation are made from the least cost

The starting solution is

$$x_{12} = 15, x_{23} = 5, x_{24} = 10, x_{31} = 5, x_{34} = 5$$

The associative objective value is

$$Z = 2 \times 15 + 5 \times 4 + 9 \times 15 + 18 \times 5 + 10 \times 20$$

$$Z = 475$$

Teacher's Signature \_\_\_\_\_

AM

Page No. \_\_\_\_\_

Q MR Autos has three plants in Los Angeles, Detroit and New Orleans and two distribution centres in Denver and Miami. The quarterly capacities at three plants are 1000, 1500 and 1200 cars and the demands at two distribution centres are for the same period are 2300 and 1400 cars. The mileage chart between the plants and the distribution centres

|             | Denver | Miami | Supply          |
|-------------|--------|-------|-----------------|
| Los Angeles | 80     | 215   | <del>1000</del> |
| Detroit     | 100    | 108   | <del>1500</del> |
| New Orleans | 102    | 68    | <del>1200</del> |
| Demand      | 2300   | 1400  |                 |

Solution

|             | Denver             | Miami       | Supply              |
|-------------|--------------------|-------------|---------------------|
| Los Angeles | 80<br>1000<br>1300 | 215<br>200  | <del>1000</del>     |
| Detroit     | 100                | 108<br>1200 | <del>1500</del> 200 |
| New Orleans | 102                | 68          | <del>1200</del>     |
| Demand      | 2300<br>1300       | 1400<br>200 |                     |

Teacher's Signature \_\_\_\_\_

AMP

Page No.

The arrow shows the order in which allocations are made from least cost

The associative value is

$$Z = (68 \times 1000) + (1000 \times 80) + (100 \times 1200) + (108 \times 200)$$

$$Z = \$ 313,200$$